

Evaluation of Machine Learning Approaches for Cirrhosis Survival Prediction

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Abstract—Cirrhosis is a fatal condition characterized by irreversible scarring and loss of liver function. Prediction of cirrhosis treatment outcomes at an early and precise manner is difficult because of issues like data imbalance, feature dimensionality, noises and outliers prevalent in clinical dataset. This study examines the use of supervised machine learning approaches to support this prediction in cirrhosis patients. We analyze the performance of seven predictive models, such as Decision Trees, Random Forests, Gradient Boosting Machines (GBMs), K-Nearest Neighbors, Extra Trees, and Logistic Regression based on their accuracy, F-1 score, Recall, Precision, and Matthews Correlation Coefficient (MCC). Outlier detection using isolation forests is used to refine data quality, while SMOTE resampling addresses class imbalance and a Genetic Algorithm optimizes feature selection for improved model performance. LightGBM and XGBoost proved to be the most successful methods with an AUC of 0.94. The combination of XG-Boost with Genetic Algorithms demonstrates superior predictive power, marking a novel methodology for automating liver disease diagnosis. The results show that the combination of the XGBoost algorithm and Genetic Algorithms is best suited for automating liver disease diagnosis. This methodology promotes early diagnosis and intervention based on improvements in the efficiency and precision of predictions.

Index Terms—Keywords: Liver Cirrhosis; Health, Machine Learning, Random Forest, Extra Trees, Logistic Regression Histogram Gradient Boosting, LightGBM, XGBoost, KNN, CatBoost

I. INTRODUCTION

The liver, the largest organ in the human digestive system vital for digestion, also is primarily responsible for detoxification, metabolism and Bilirubin clearance[1]. All diseases of the liver reach their most dangerous forms in cirrhosis, as it contributes to permanent scarring of the tissue of the liver. Chronic alcoholism, viral infections including hepatitis B and C, and the steadily growing incidence of non-alcoholic fatty liver disease contribute to cirrhosis. Liver diseases account for between 2 million and more deaths per year, and nearly a million die each year from cirrhosis. The disease ranks as the 11th leading cause of death worldwide [2].

Early symptoms of cirrhosis usually are asymptomatic and go unnoticed by the patient until the disease has fully manifested. Traditionally, scoring systems to predict the prognosis[3] in this group of patients were used: the Child-Turcotte-Pugh score and the Model for End-Stage Liver Disease score. However, these scoring systems have limitations in the accuracy and objectivity of long-term survival predictions across the spectrum of liver disease severity [4].

Although machine learning has received extensive applications to enhance the predictive accuracy in various domains of health, scarce research studies have been conducted in the form of specific machine learning algorithms for cirrhosis survival outcome prediction. To bridge that gap, this study compares the performances of a wide range of machine learning algorithms, which include Logistic Regression, K- Nearest Neighbours, Random Forest, Extra Trees Classifier, Histogram-Based Gradient Boosting, Light Gradient Boosting Machine, Extreme Gradient Boosting, and Categorical Boosting [5]. These classifiers are known to have excelled in doing classification tasks and will be evaluated for the comparative strength of the power to predict long-term survival in cirrhosis patients.

Such sophisticated machine learning methods would thus be presented to increase the accuracy of the forecast for maximum clinical decision-making and correct allocation of health resources. The paper is new because more accurate predictions can be provided, which is made possible due to clinical data associated with preoperative events, thus enabling early diagnosis and intervention. Indeed, other researchers claim that the introduction of machine learning in medical data analytics will improve patient outcomes, especially in patients suffering from cirrhosis [6].

II. LITERATURE REVIEW

Traditionally, the detection and prognosis of liver cirrhosis depend on techniques that could include liver biopsies, liver function tests, and imaging. All are useful methods but often invasive and prone to sensitivities and precisions that might give inaccurate detection. Over the last few years, strong alternatives based on machine learning techniques have started to gain ground with non-invasive, data-driven approaches beneficial for analyzing large-scale datasets that might otherwise be invisible to the human eye in patterns about cirrhosis patients.

Several research studies quantitatively estimated usage of machine learning algorithms in the detection and prediction of liver disease. In [7], the authors presented a comparative analysis of machine learning techniques for patients with Indian liver disease. In the paper, authors compared and contrasted algorithms like SVM, Random Forest, and Decision Trees and

concluded that the accuracy of machine learning models greatly outperformed the conventional diagnostic techniques [8]. Such results therefore clearly indicate the growing importance that liver disease diagnosis is giving to machine learning, especially among diverse populations.

Reference [9] presents significant work that the authors conducted while reporting on the use of machine learning algorithms to analyze and predict liver cirrhosis. In their case, the authors make fair use of some quite advanced algorithms such as Random Forest and Gradient Boosting. In step with current studies on liver cirrhosis, where the challenge presented is more on early detection because this disease is asymptomatic during the early stages of development. The development of the AI system through various algorithms in machine learning, specifically Random Forest, which proved to have a high percentage of accuracy of about 98%, gives physicians much ground for early diagnosis of the disease. The comparative analysis from the two studies also brought out the fact that even advanced algorithms can contribute highly to improving predictive accuracy in medical diagnostics and making more reliable diagnoses, paving the way for further advancements in this sphere.

Another significant contribution is [10], which explores the application of machine learning for the automated diagnosis of liver diseases.[11] The authors propose an enhanced XGBoost classifier combined with a genetic algorithm to improve prediction accuracy. Their study demonstrates that the Random Forest achieves the highest accuracy of 88%, outperforming traditional models such as the XGBoost and LightGBM, which yield accuracies of 86% and 86%, respectively. Additionally, the implementation of outlier detection methods, like Isolation Forest, improves data quality and reduces classification time, supporting the efficacy of advanced algorithms in medical diagnostics.

The review in [12] computer-assisted diagnosis of hepatic lesions is focused on the fact that non-invasive detection methodologies are relevant for chronic liver diseases. The preprocessing methods denoising, deblurring, and segmentation are also covered. Indeed, the features that are characterized for use in support vector machines for classification often used are texture. Said networks based on methodologies of deep learning hold a great promise, and the addition of biopsy specimens and pathological variables may add to the predictive precision. They are confident that advancements in machine learning will bridge limitations in data and bring improvements in the results of diagnosis for liver conditions.

The study in [13] mentions the significant difficulty of diagnosing liver disease, a major global health problem affected by factors such as obesity and liver disease. It aims to improve accurate diagnosis while reducing the cost by evaluating the performance of various machine learning algorithms such as logistic regression, K-nearest neighbors, decision trees, support vector machines, naive Bayes, and random forests. Studies have shown that logistic regression has the highest accuracy of 75%, followed by random forest with 74%, demonstrating the ability

of machine learning to improve pain diagnosis from a good predictive model [13].

Finally, [14] addressed the potential possibility of Explainable AI (XAI) for Enhanced Interpretation of Liver Cirrhosis Biomarkers. It was conducted with an objective of increasing the interpretability and transparency of machine learning approaches applied for cirrhosis detection. The application of XAI techniques for tree-based models showed that interpretable machine learning becomes of paramount importance for increasing both trustworthiness and acceptability of AI-driven solutions in clinical practice.

III. PROPOSED FRAMEWORK

The primary objective of this research is to propose a methodology for developing predictive models for liver disease using a range of supervised machine learning algorithms. A comparative analysis of the proposed approach has been conducted, with performance evaluated through various classification metrics. A concise overview of each step involved in the disease prediction process is outlined as follows [10] -

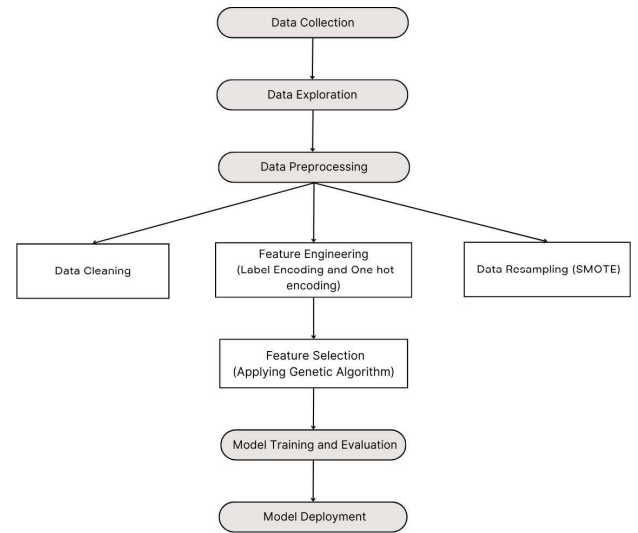


Fig. 1. Schematic Representation of Proposed Methodology

A. Data Collection

The data set that is taken into consideration for the following paper is the Cirrhosis Patient Survival Prediction set from UCI [15][10]. Table 1 presents a dataset of 418 patients with liver cirrhosis, including 20 clinical factors that may be used to forecast their survival.

This dataset includes a target attribute referred to as "Status," which represents the patient's health outcome over time. Specifically, "Status" is a categorical variable, utilized in a multi-class classification framework, and is categorized into three distinct states: C (censored, indicating the patient was alive at the last follow-up), CL (censored due to liver transplantation), and D (indicating the patient's death).

TABLE I DaTasET

Index	Attribute	Range
1	ID	1-418
2	N DAYS	41-4795
3	Status	C/CL/CD
4	Drug	D-penicillamine/Placebo
5	Age(Days)	9598-28650
6	Sex	F/M
7	Ascites	Y/N
8	Hepatomegaly	Y/N
9	Spiders	Y/N
10	Edema	Y/N/S
11	Bilirubin	0.30-28
12	Cholesterol	120-1775
13	Albumin	1.96-4.64
14	Copper	4-588
15	Alk Phos	289-13862.40
16	SGOT	26.35-457.25
17	Tryglicerides	33-598
18	Platelets	62-721
19	Prothrombin	9-18
20	Stage	1-4

B. Data Exploration

This process enables a deeper understanding of the data's structure and potential trends before moving to model building. During data exploration, various visualization tools, such as heatmaps, boxplots, and dependency plots, are used to analyze feature interactions and identify any extreme values or outliers that may affect the analysis. Specifically, heatmaps allow for a detailed study of correlations between features, highlighting attributes that may be highly linearly dependent. This step is essential in guiding subsequent data preprocessing and feature selection, ensuring a more accurate predictive model.

C. Data Preprocessing

- *One-hot Encoding* - In using this method, data is preprocessed first so that any categorical variable can be encoded for readability with machine-learning algorithms. For "Edema" attribute, one-hot encoding was particularly used in the project.
- *Label encoding for categorical* - This type of preprocessing over data in which categorical variables are converted to numerical format by assigning an integer to each category is called label encoding. In contrast, one-hot encoding creates more than one binary column for the categories. In label encoding, categories can be represented in a single column with each unique category appearing as an integer. This is helpful when the categorical variable contains a natural ranking and/or even large amounts of categories to reduce the number of dimensions along with saving memory.
- *Elimination of Duplicate Values* - To improve data quality and analytical efficiency, redundant values are removed from the dataset.
- *Resampling* – Given the data imbalance, where patients with liver disease are in the majority and those without are in the minority, the Synthetic Minority Over-sampling Technique (SMOTE) is applied.

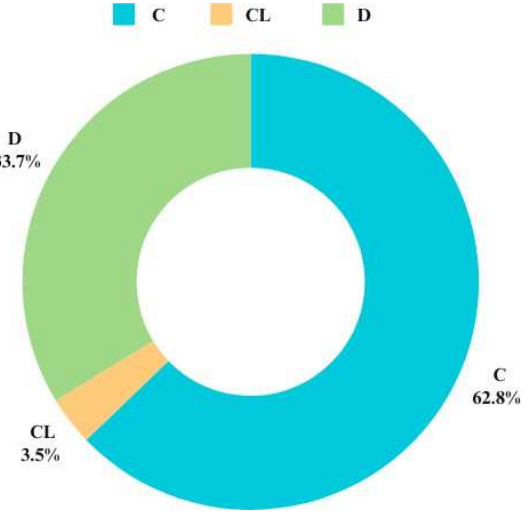


Fig. 2. Class Distribution of Target Attribute - Status

- *Outlier Detection and Elimination* – Outliers, or extreme values that deviate significantly from the dataset's general pattern, often result from measurement errors or inconsistencies. For multivariate outlier detection, the Isolation Forest algorithm is used to identify and remove contaminated data points [10].
- *Handling Noisy Data* - To handle missing values in a dataset, we can employ a combination of imputation techniques. For numerical features, the KNNImputer is effective as it replaces missing values by averaging the values of the k-nearest neighbors for each sample. However, since KNNImputer does not cater to categorical data, we can use SimpleImputer with the 'most_frequent' strategy for those columns, which substitutes missing values with the most common value in each category.

D. Feature Selection

In reality, uncovering pertinent features is a key element in machine learning, especially in the area of medical diagnostics, which is one way to achieve the accuracy of the model and eliminate the difficulty. The Genetic Algorithm (GA) provides a very successful tool for this, which aims at mimicking the biology by choosing the subset of the most important features. It begins with a population of arbitrary parents whose children, being unicorns, are then evaluated using a fitness function based on model performance metrics. The systematic step allows the GA to dribble to a cadre that gives back information of the data with maximum precision and minimum overfitting.

E. Overview of Classification Algorithms Used

Various machine learning algorithms are employed for classification, such as -

- *Logistic Regression* - is well-suited for multi-class classification tasks where the target variable is categorical, allowing it to model complex class boundaries even in imbalanced datasets.
- *K-Nearest Neighbors* - operates on proximity of similar data points. It classifies a data point by examining the classes of its nearest neighbors in the feature space, making it effective for tasks where the relationships among data points are crucial [16].
- *Extra Trees (Extremely Randomized Trees)* - classifier is an ensemble learning method that constructs multiple uncorrelated decision trees using random subsets of data features [17]. By randomizing the split points within each feature, it reduces variance and combats overfitting, making it well-suited for multi-class classification on imbalanced datasets with high-dimensional feature spaces.
- *Random Forest* - is an ensemble learning method widely used for classification tasks, where several decision trees are built on randomly selected data subsets and integrated to improve model accuracy. Each tree makes its prediction, with the final output derived from either a majority vote or averaging, depending on the nature of the problem. This strategy boosts the model's resilience, minimizes overfitting, and generally achieves stronger generalization compared to using a single decision tree
- *Histogram-based Gradient Boosting (HistGBM)* - is a specialized form of gradient boosting that accelerates training by grouping continuous data into discrete bins, thus reducing computational costs. This algorithm is effective for imbalanced, multiclass datasets with numerous features, as it can handle complex patterns and large data volumes efficiently, offering both speed and accuracy improvements.
- *XGBoost* - is an optimized gradient boosting algorithm that enhances performance through parallel processing. It incorporates regularization techniques to combat overfitting, making it an excellent option for handling complex classification tasks.
- *Light Gradient-Boosting Machine* - is a high-performance algorithm that employs decision tree methodologies [17]. Unlike traditional approaches, it splits trees leaf-wise rather than depth-wise, which can lead to more efficient training and improved model performance.
- *Categorical Boosting* - is a gradient boosting algorithm that excels in handling categorical features and reduces overfitting through ordered boosting. It is particularly advantageous for imbalanced, multiclass problems as it can natively process categorical data and provides robust performance without extensive preprocessing, enhancing predictive accuracy and efficiency in high-dimensional datasets.

F. Metrics for Performance Evaluation

Since it is a multiclass prediction, we proceeded with the usual evaluation metrics, namely accuracy, precision, recall, F1 score, MCC, ROC curve, and confusion matrix.

- **Accuracy:** This metric measures the percentage of correctly predicted instances over total instances. This represents the overall classifier performance.

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (1)$$

- **Precision:** This metric calculates the ratio of correctly identified positive cases to all cases that were predicted as positive.

$$Precision = \frac{TP}{TP+FP} \quad (2)$$

- **Recall :** This metric evaluates the ability of the model to correctly predict positive instances. This metric is vital for understanding how well the model identifies positive cases, especially in critical applications like medical diagnoses [8].
- **F1 Score:** represents the relationship between the actual and the yield by providing an equal measure that takes into account the negative and the negative. This measure is especially useful when the class distribution is unequal.
- **MCC** is a metric used to evaluate model performance, especially in multiclass and imbalanced datasets. It combines all four confusion matrix categories into a single score from -1 to +1, where +1 indicates perfect predictions, 0 random predictions, and -1 total disagreement

$$MCC = \sqrt{\frac{(TP \times TN) - (FP \times FN)}{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (3)$$

- **ROC Curve:** The ROC curve graphically illustrates the balance between the true positive rate and the false positive rate at various threshold settings. This metric is crucial for evaluating the model's performance in distinguishing between classes, especially in imbalanced datasets [19]. A higher area under the ROC curve (AUC) indicates better classification capability [18].
- **Confusion Matrix:** By displaying predictions for each class, the confusion matrix uncovers any frequent misclassifications, allowing for targeted model improvements. This analysis can suggest modifications, like fine-tuning the model structure or increasing training data for certain categories. This metric is essential for assessing model performance, as it helps calculate accuracy, precision, and recall while identifying specific misclassification types.

IV. RESULTS

The evaluation of multiple machine learning models reveals consistently high performance across the board, with all models achieving accuracy scores above 0.75. XGBoost and LightGBM stand out with the highest accuracy scores of 0.84 and an AUC of 0.94, highlighting their superiority in handling complex data patterns. These algorithms leverage advanced boosting techniques, which iteratively refine predictions by emphasizing harder-to-predict cases. Additionally, both algorithms efficiently handle

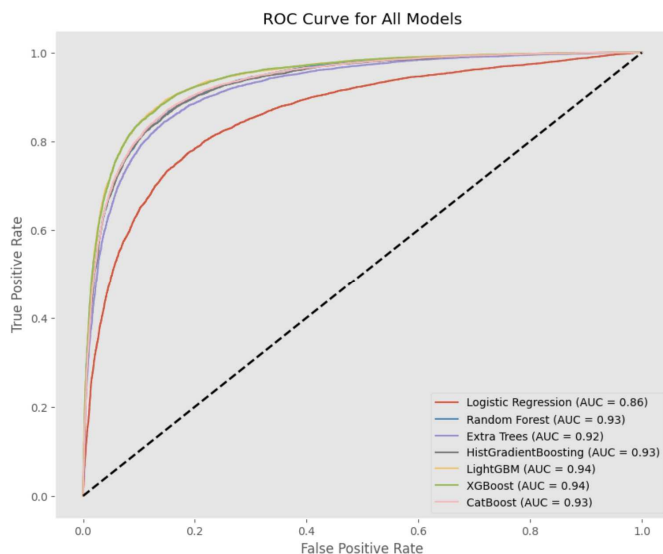


Fig. 3. Receiver Operating Characteristics (ROC) Curve

high-dimensional datasets, and their ability to use parallelization accelerates computations without compromising accuracy. This efficiency is particularly advantageous in scenarios with large feature spaces or imbalanced datasets, as seen in our study. HistGBM, Random Forest, and CatBoost also have acceptable

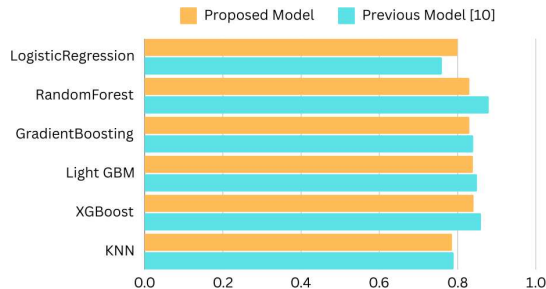


Fig. 4. Accuracy Comparison

AUC values, which are around 0.93. The use of a histogram for partitioning of continuous features in HistGBM makes it less computationally intensive and requires less memory than CART. One of the boosting algorithms is Random Forest, which combines many decision trees, this makes it more resistant to overfitting compared to boosting but may not have the fine-tuned Recognizable Pattern (R-Pattern) boosting capability when working with intricate data. The conclusion can be made that CatBoost has a high quality of native handling of categorical features, but its performance can also be rather highly dependent on the correct tuning of hyperparameters and pre-processing of the input data.

Logistic Regression, which gave an AUC of 0.86 is again slightly inferior to the more complex models. It is a linear classifier and thus has the limitation of not being able to model

complex, nonlinear patterns in the data. This limitation helps explain why it performs less well than other ensemble methods like XGBoost and the Random Forest, which are well-suited for this type of relationship.

The Matthews Correlation Coefficient (MCC) scores are also highly reliable and range from 0.66 at the highest. MCC shows a fair assessment of true and false positives/negatives that demonstrates the models' ability to generalize across imbalanced data distribution. Overall, the improvements made to Logistic Regression, while not very high, show the effectiveness of feature selection based on Genetic Algorithms as well as the positive effect of SMOTE resampling, which can reduce the impact of presence of imbalance and outliers.

All in all, it appears that boosting algorithms are the most suitable for this problem since they balance different data characteristics and are designed to work with them. However, compared to other methods such as the Logistic Regression and Random Forest, there is slight compromise in the effectiveness in these tasks. These results highlight the roles of algorithm selection, data preprocessing, and feature engineering in the diagnosis of liver disease based on predictive modeling.

TABLE II
EVALUATION METRICS OF DIFFERENT ML ALGORITHMS

Algorithm	Accuracy	Precision	Recall	F1	MCC
Logistic Regression	0.8	0.77	0.8	0.78	0.58
Random Forest	0.83	0.83	0.83	0.82	0.64
Extra Trees	0.82	0.81	0.82	0.8	0.61
Histogram GBM	0.83	0.82	0.83	0.82	0.64
Light GBM	0.83	0.83	0.83	0.83	0.66
XGBoost	0.84	0.83	0.84	0.83	0.66
CatBoost	0.83	0.82	0.83	0.82	0.64
KNN	0.78	0.77	0.78	0.77	0.54

V. CONCLUSION

In this study, we explored the forecasting ability of tree-based machine learning algorithms—Random Forest, Extra Trees, HistGradientBoosting, LightGBM, XGBoost, and CatBoost—to enhance the prognosis of cirrhosis in the survival situation.

The impact of this research is two-sided: on one hand, the higher accuracy model enables a more streamlined resource allocation and patient care in the medical field, on the other hand, it exemplifies machine learning as a powerful tool for diagnostic support that can be customized to the constantly-changing clinical picture. By employing the combination of these sophisticated software, healthcare providers will be in a better position to prognosticate disease progression and modify treatment ideas, which will lead to a decrease in cirrhosis-related mortality and hence less burden to the healthcare system. Further studies should be done to add to these findings; for example, hybrid and deep learning models should be explored, and datasets should be used more freely to refine the prediction of cirrhosis. The main thrust of our research is to underscore the potential of data-driven methods in shaking up the traditional liver disease prognosis register. Consequently, this will be a milestone in the development of data-centred personalized healthcare.

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